Lectures 17/18

Math 4741: Mathematical Analysis

Chapter 9: Analysis of Networking Protocols

Texts: Analysis of Computer and Communication Networks – Fayez Gebali (Chapter 10)

Modeling Medium Access Control Protocols

Introduction

Pure ALOHA
Slotted ALOHA
IEEE Standard 802.3: carrier sense multiple access with collision detection (CSMA/CD)
Carrier sense multiple access with collision avoidance (CSMA/CA)

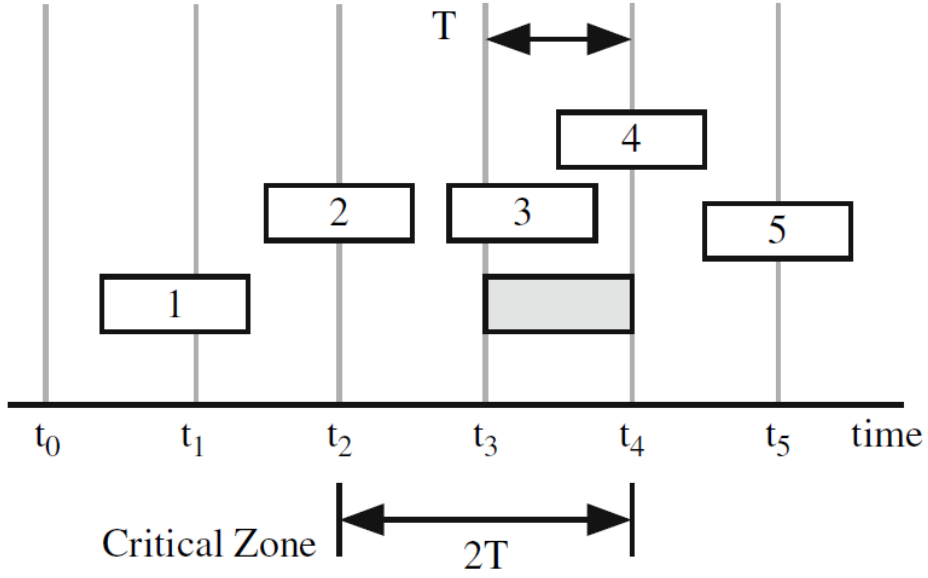
Performance Analysis of ALOHA

- User with a frame transmits immediately
- Two users transmitting simultaneously resulting in a collision
- Collision is detected by listening the medium (wired) or by the absence of an ACK (wireless)

Assumptions

- Frames have equal lengths
- Each frame has a duration of T
  - T is maximum transmission delay between any pair of STAs
- Conflict: how the transmission of a frame is affected by other frames
- Critical Time: Time period in which transmission from others will result in a collision
  - Any frame transmitted in the period T before the transmission of the tagged frame
  - Any frame transmitted during the transmission time T of the tagged frame

Critical Time



Frame	Start time t	Contention with shaded frame
1	$t_0 < t < t_1$	No
2	$t_1 < t < t_2$	No
3	$t_2 < t < t_3$	Yes
4	$t_3 < t < t_4$	Yes
5	$t_4 < t < t_5$	No

Markov Models of ALOHA

1. The states of the Markov chain represent the status of the wireless channel: *idle*, *transmitting*, and *collided*.
2. The propagation delay between any pair of users is less than the frame time T.
3. The time step value T is taken equal to the frame transmission delay.
4. There are N users in the system.
5. Users can transmit any time they want.
6. The probability that a user transmits a frame in one time step is a.
7. All frames have equal lengths and the duration of each frame is T.
8. Contention occurs if a frame is sent at time t and there are transmissions during the time period $T - t$ to $t + T$.
9. A user retransmits a corrupted frame after waiting a random amount of time.

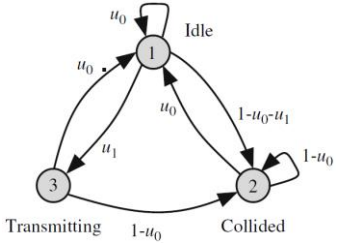
Markov Models of ALOHA

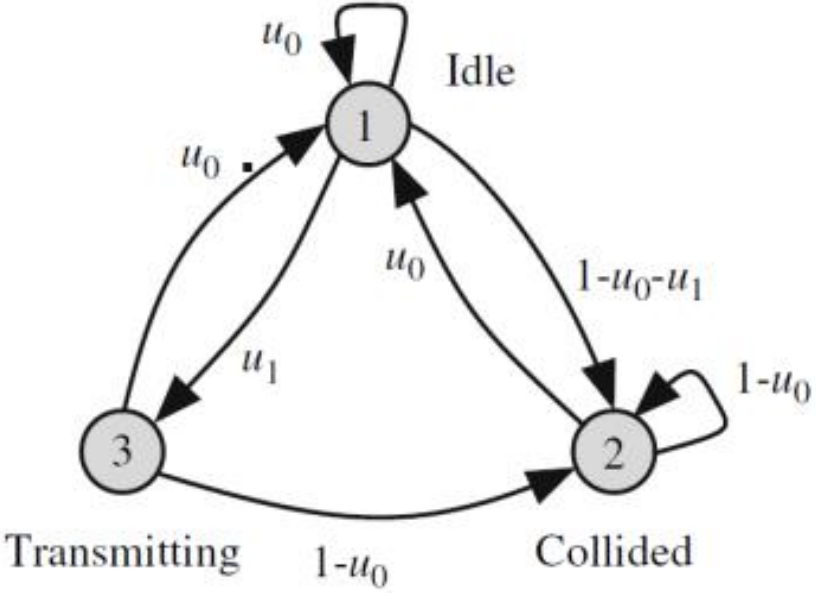
Based on the assumptions, the channel can be in one of three states:

Idle state: We remain in s_1 as long as all users are idle (probability u_0). We move to transmitting state if exactly one user requests access (probability u_1) and we move to collided state if two or more users request access (probability $1 - u_0 - u_1$).

Transmitting state: We move to idle state if all users are idle. If one or more users request access, we move to collided state since there will be no period of calm before the next transmission.

Collided state: We move to idle state if all users are idle. Any arriving requests keep the system in collided state.




u_0
Idle
1
u_0
u_0
$1-u_0-u_1$
u_1
$1-u_0$
3
2
Transmitting
$1-u_0$
Collided

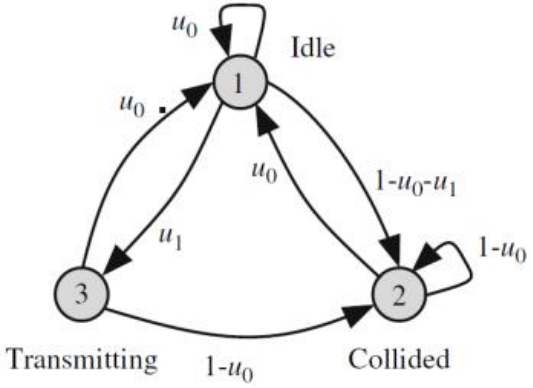
The probability that k users request access during a time step is given by binomial statistics:

$$p_k = \binom{N}{k} a^k (1-a)^{N-k} \tag{10.14}$$

$$u_0 = (1-a)^N$$

$$u_1 = N\, a\, (1-a)^{N-1}$$

Probability Transition Matrix



$$P = \begin{bmatrix} u_0 & 1-u_0-u_1 & u_1 \\ u_0 & 1-u_0 & 0 \\ u_0 & 1-u_0 & 0 \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} u_0 & u_0 & u_0 \\ 1-u_0-u_1 & 1-u_0 & 1-u_0 \\ u_1 & 0 & 0 \end{bmatrix}$$

Steady-stare Probabilities

Let the steady-state probability vector $\pi = [\pi_1\ \pi_2\ \pi_3]$

The system of linear equations

$$\pi_1 = u_0\pi_1 + u_0\pi_2 + u_0\pi_3$$

$$\pi_2 = (1 - u_0 - u_1)\pi_1 + (1 - u_0)\pi_2 + (1 - u_0)\pi_3$$

$$\pi_3 = u_1\pi_1$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

Steady-state Probabilities

The solution to the system of linear equations is

$$\pi_1 = u_0$$

$$\pi_2 = 1 - u_0 - u_0 u_1$$

$$\pi_3 = u_0 u_1$$

throughput of ALOHA, which is given by

$$\text{Th} = s_3 = u_0\, u_1$$

Substituting the values of u_0 and u_1 from (10.14), we get

$$\text{Th} = N\, a\, (1 - a)^{2N-1}$$

For large N, the throughput can be expressed

$$\text{Th} = Nae^{-2Na}$$

The throughput is measured in units of frames/time step.

The throughput in units of frames/s is

$$\text{Th}' = \frac{\text{Th}}{T}$$

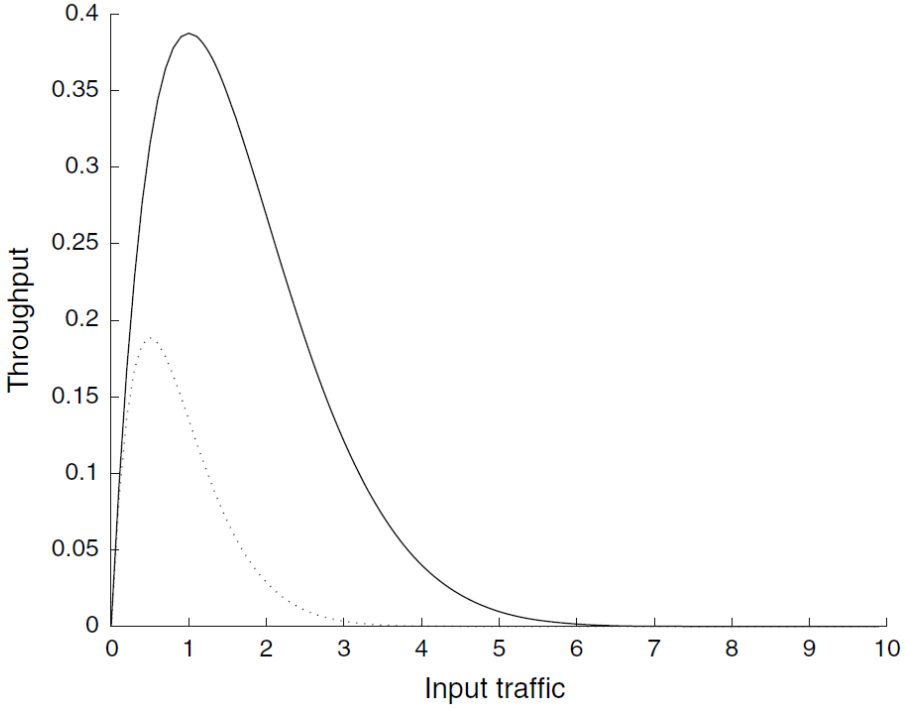


Fig. 10.4 The throughput for ALOHA versus the average number of requests per time step for the case $N = 10$. *Dotted line* is for ALOHA and *solid line* is for slotted ALOHA

The input traffic is found from the binomial distribution

$$N_a(\text{in}) = N\ a$$

We define the *acceptance probability* or *efficiency* p_a as the probability of a successful transmission for a given user. p_a is given by

$$\begin{aligned} p_a &= \frac{\text{Th}}{N_a(\text{in})} \\ &= (1-a)^{2N-1} && (10.24) \\ &\approx \mathrm{e}^{-2aN} && (10.25) \end{aligned}$$

The last equation relies on the approximation

$$\lim_{n\to\infty} \left(1 - \frac{x}{n}\right)^n = \mathrm{e}^n$$

The average number of attempts before a successful transmission is

$$n_a = \sum_{n=0}^{\infty} n \ (1 - p_a)^n \ p_a$$

This evaluates to

$$n_a = \frac{1 - p_a}{p_a}$$

If the average duration of the random wait period is τ (in seconds), then the average wait for a frame before successful transmission is

$$W = n_a \times \tau = \frac{\tau \, (1 - p_a)}{p_a} \qquad \text{seconds} \qquad (10.31)$$

Assume that the number of stations is fixed. In that case, we can vary the arrival probability a and investigate how the throughput varies with input traffic. The maximum throughput occurs when

$$\frac{\mathrm{d}\,\mathrm{Th}}{\mathrm{d}\,a} = 0 \tag{10.32}$$

Differentiating (10.20) indicates that the maximum throughput for ALOHA occurs when

$$a_0 = \frac{1}{2\,N} \tag{10.33}$$

Thus, the maximum throughput is theoretically equal to

$$\mathrm{Th(max)} = \frac{1}{2}\left(1 - \frac{1}{2N}\right)^{2N-1} \tag{10.34}$$

The above equation gives the maximum throughput for any number of users N. A simple expression is obtained when the user population is very large, $N \to \infty$:

$$\begin{aligned} \mathrm{Th(max)} &= \frac{1}{2\,\mathrm{e}} \\ &= 18.394\% \end{aligned} \tag{10.35}$$

Performance Analysis of slotted ALOHA

Assumptions

Slotted ALOHA was proposed to improve the throughput of the original ALOHA [2]. As the name implies, time in slotted ALOHA is divided into slots and the value of one time step equals the frame time T. Users know about the start of a new slot through a synchronizing signal that is transmitted by a source. A user with data to send is permitted to transmit only at the start of a time step. This removes the chaos of pure ALOHA and improves the efficiency as we shall see below.

Markov Model for Slotted ALOHA

Based on our assumptions, the wireless channel can be in one of three states: *idle*, *collided*, or *transmitting*. Figure 10.5 shows the transition diagram for our Markov chain. Further, the time step value is naturally chosen equal to the slot time value. Now is a good time for the reader to compare that figure with the pure ALOHA transition diagram of Fig. 10.3. What is the major difference?

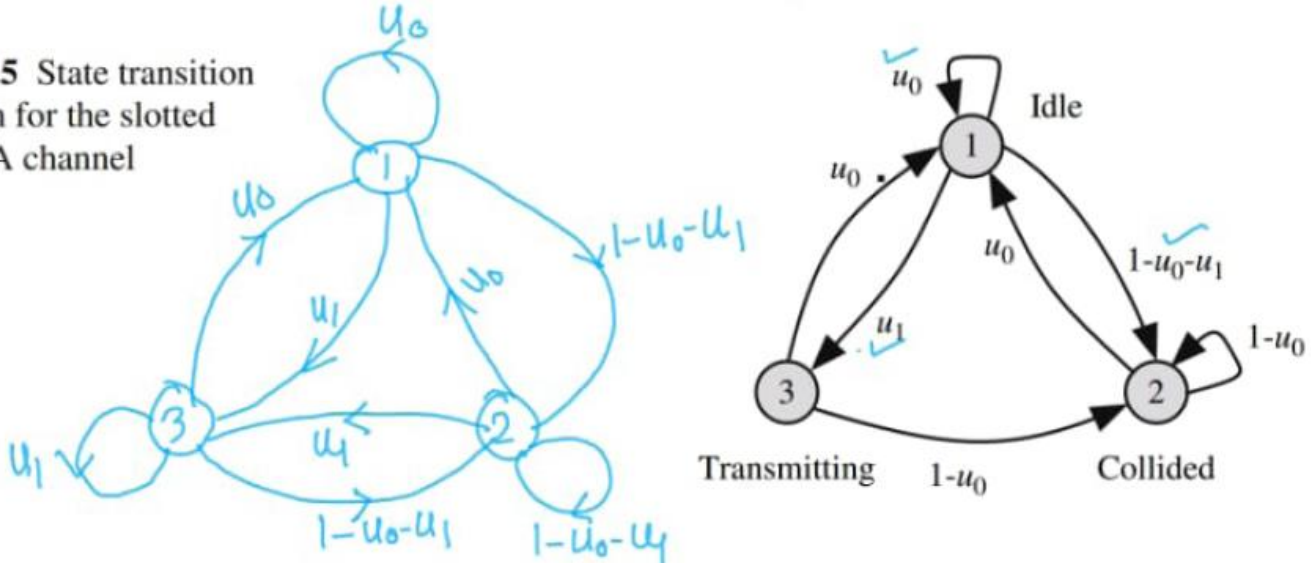


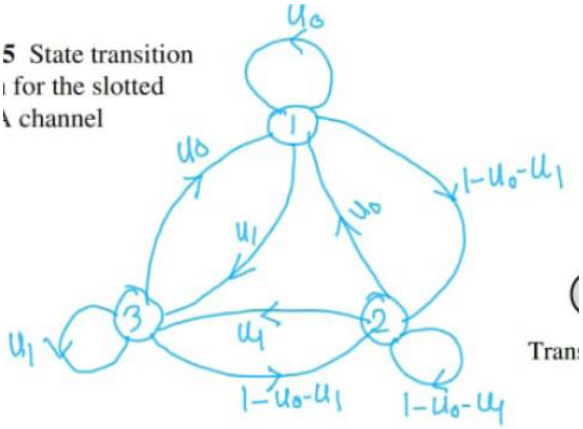
Fig. 10.5 State transition diagram for the slotted ALOHA channel

Idle state: We remain in s_1 as long as all users are idle (probability u_0). We move to transmitting state if exactly one user requests access (probability u_1) and we move to collided state if two or more users request access (probability $1 - u_0 - u_1$).

Transmitting state: We move to idle state if all users are idle. We move to transmitting state if exactly one user requests access (probability u_1) and we move to collided state if two or more users request access (probability $1 - u_0 - u_1$).

Collided state: We move to idle state if all users are idle. We move to transmitting state if exactly one user requests access (probability u_1) and we stay in collided state if two or more users request access (probability $1 - u_0 - u_1$).

5 State transition
for the slotted
channel



Tran

The probability that k users request access during a time step is given as before by binomial statistics:

$$p_k = \binom{N}{k} a^k (1-a)^{N-k} \tag{10.40}$$

The transition matrix of the system is

$$\mathbf{P} = \begin{bmatrix} u_0 & u_0 & u_0 \\ 1-u_0-u_1 & 1-u_0-u_1 & 1-u_0-u_1 \\ u_1 & u_1 & u_1 \end{bmatrix} \tag{10.41}$$

At equilibrium, the distribution vector is obtained by solving the two equations

$$\mathbf{P}\,\mathbf{s} = \mathbf{s} \tag{10.42}$$

$$s_1 + s_2 + s_3 = 1 \tag{10.43}$$

The solution to the above two equations is simple:

$$\begin{aligned} s_1 &= u_0 \\ s_2 &= 1 - u_0 - u_1 \\ s_3 &= u_1 \end{aligned} \tag{10.44}$$

Comparison between Pure ALOHA and Slotted ALOHA

1. Pure ALOHA and slotted ALOHA channels have the same probability of being in state s_1 (the idle state).
2. Slotted ALOHA has a *lower* probability of being in state s_2 (the collided state) compared to pure ALOHA.
3. Slotted ALOHA has a *higher* probability of being in state s_3 (the transmitting state) compared to pure ALOHA.

All these factors contribute to enhance the performance of slotted ALOHA.

Slotted ALOHA Performance

We are interested in the throughput of slotted ALOHA, which is given by

$$\text{Th} = s_3 = u_1 = N\ a\ (1-a)^{N-1} \tag{10.45}$$

A frame will be transmitted if only one user requests to access the channel irrespective of the previous state of the channel. For large N, the throughput can be expressed in terms of the input traffic as

$$\text{Th} = Nae^{-Na} \tag{10.46}$$

The throughput in units of frames/s is

$$\text{Th}' = \frac{\text{Th}}{T}$$

The average number of users is given by the binomial distribution

$$N_a(\text{in}) = N\ a$$

We define the *acceptance probability* or *efficiency* p_a as the probability of a successful transmission for a given user. p_a is given by

$$\begin{aligned} p_a &= \frac{\text{Th}}{N_a(\text{in})} \\ &= (1-a)^{N-1} && (10.49) \\ &\approx \mathrm{e}^{-aN} && (10.50) \end{aligned}$$

The average number of attempts before a successful transmission is

$$n_a = \sum_{n=0}^{\infty} n \ (1 - p_a)^n \ p_a \tag{10.54}$$

This evaluates to

$$n_a = \frac{1 - p_a}{p_a} \tag{10.55}$$

If the average duration of the random wait period is τ (in seconds), then the average wait for a frame before successful transmission is

$$W = n_a \tau = \frac{\tau \, (1 - p_a)}{p_a} \qquad \text{seconds} \tag{10.56}$$

Assume that the number of stations is fixed. In that case, we can vary the arrival probability a and investigate how the throughput varies. The maximum throughput occurs when

$$\frac{\mathrm{d}\,\mathrm{Th}}{\mathrm{d}\,a} = 0 \tag{10.57}$$

Differentiating (10.20) indicates that the maximum throughput for ALOHA occurs when

$$a_0 = \frac{1}{N} \tag{10.58}$$

Maximum Throughput of Slotted ALOHA

Thus, the maximum throughput is theoretically equal to

$$\text{Th(max)} = \left(1 - \frac{1}{N}\right)^{N-1} \tag{10.59}$$

The above equation gives the maximum throughput for any number of users N. A simple expression is obtained when the user population is very large, $N \to \infty$:

$$\begin{aligned} \text{Th(max)} &= \frac{1}{\text{e}} \\ &= 36.788\% \end{aligned} \tag{10.60}$$

Performance Analysis of CSMA/CD (IEEE 802.3)

The IEEE 802.3 standard is used for *wired LANs* where the time required for *one bit* to travel between the two farthest stations (propagation time) is much smaller than the time required for *one frame* to be sent by the sender (transmission delay).

Signals on the channel travel very close to the speed of light and it takes a finite amount of time before all stations become aware that a channel is starting to access the medium. Therefore, a collision is said to take place when two or more stations start transmitting within the frame propagation delay. Because during this time, transmitting stations think that the medium is idle. When that happens, the two colliding stations *stop transmitting* and wait for a random amount of time before attempting to transmit again. This reduces the chance that the stations will once again

At this point, it is worthwhile mentioning three different CSMA access strategies:

- 1-persistent CSMA
- Nonpersistent CSMA
- p-persistent CSMA

In a 1-persistent CSMA, a station with frame to send senses the channel. If the channel is sensed free, the frame is sent. If the channel is busy, the station *continuously* monitors the channel and sends the frame when the channel is sensed idle.

In a nonpersistent CSMA, a station with frame to send senses the channel. If the channel is sensed free, the frame is sent. If the channel is busy, the station *waits* for a random amount of time before monitoring the channel and sends the frame when the channel is sensed idle. If the channel is sensed busy, the station repeats the random wait.

In a p-persistent CSMA, a station with frame to send senses the channel. If the channel is sensed free, the frame is sent with probability p. The transmission is deferred for the next time slot with probability $1 - p$. This process is repeated until the frame is sent.

Model Assumptions

1. Since the current state of the channel depends only on its immediate past history, we can model the channel using Markov chain analysis.
2. The states of the Markov chain represent the states of the channel: *idle*, *transmitting*, and *collided*.
3. The channel is shared among N stations.
4. There is a single station class (equal priority).
5. The frame arrival probability per time step is a.
6. All transmitted frames have equal lengths.
7. A frame duration is equal to the transmission delay of a frame τ_t.
8. The time step of the Markov chain is chosen equal to the collision detection delay, i.e., $T = \tau_c$.

9. We define n as the ratio of transmission delay to propagation delay, i.e., $n = \tau_t/\tau_c$. We assume that $n > 1$, which is true for small LANs carrying long frames or operating at low transmission speeds.
10. The time required to detect a collision is less than or equal to the time step value T.
11. Probability that an idle station receives a frame for transmission during a frame transmission time (or frame duration) is a.
12. A p-persistent CSMA/CD is assumed. This is equivalent to a 1-persistent CSMA/CD with a backoff strategy.
13. An *adaptive backoff strategy* is assumed where a collided user starts to sense the channel with probability α, which is taken equal to the frame arrival probability (i.e., $\alpha = a$). In that sense, each collided user adapts its request probability α to be proportional to its incoming traffic probability a.
14. A station can have at most one message waiting for transmission.

State transition diagram/graph

The probability that i users are active at a given time step is given by

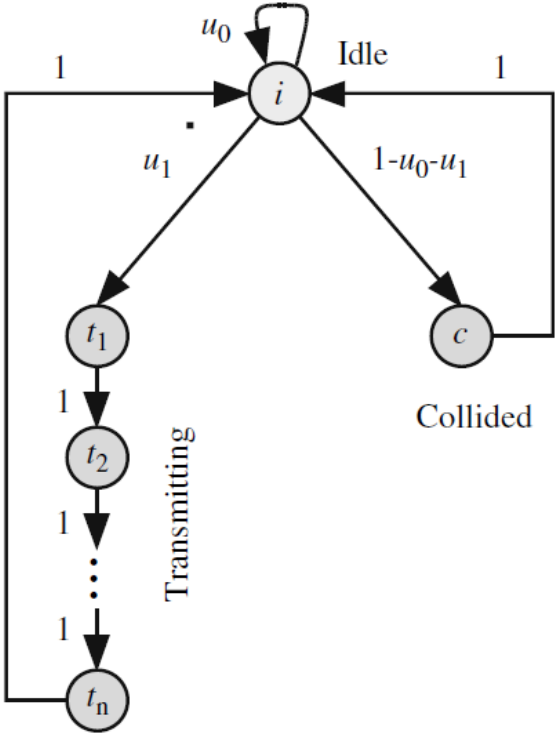
$$u_i = \binom{N}{i} a^i (1-a)^{N-i}$$

where a is the probability that a station requests a transmission during a time step.

We organize the distribution vector at equilibrium as follows:

$$\mathbf{s} = \begin{bmatrix} s_i & s_{t_1} & s_{t_2} & \cdots & s_{t_n} & s_c \end{bmatrix}^t \tag{10.66}$$

Fig. 10.6 State transition diagram for the IEEE 802.3 CSMA/CD channel



$$\mathbf{s} = \begin{bmatrix} s_i & s_{t_1} & s_{t_2} & \cdots & s_{t_n} & s_c \end{bmatrix}^t \tag{10.66}$$

The corresponding transition matrix of the channel is given by

$$\mathbf{P} = \begin{bmatrix} u_0 & 0 & 0 & 0 & \cdots & 0 & 1 & 1 \\ u_1 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ 1-u_0-u_1 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix} \tag{10.67}$$

$$p = \begin{bmatrix} u_0 & u_1 & 0 & 0 & 1-u_0-u_1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The system of linear equations for the Markov chain

$$\pi_0 = \pi_0 u_0 + \pi_3 + \pi_c$$

$$\pi_1 = \pi_0 u_1$$

$$\pi_2 = \pi_1 = \pi_0 u_1$$

$$\pi_3 = \pi_2 = \pi_0 u_1$$

$$\pi_c = \pi_0 (1 - u_0 - u_1)$$

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_c = 1$$

$$\pi_0 u_0 + \pi_0 u_1 + \pi_0(1 - u_0 - u_1) + \pi_0 u_1 + \pi_0 u_1 + \pi_0 u_1 + \pi_0(1 - u_0 - u_1) = 1$$

$$=> \pi_0 u_0 + \pi_0 u_1 + \pi_0 - \pi_0 u_0 - \pi_0 u_1 + \pi_0 u_1 + \pi_0 u_1 + \pi_0 u_1 + \pi_0 - \pi_0 u_0 - \pi_0 u_1 = 1$$

$$=> 2\pi_0 + 2\,\pi_0 u_1 - \pi_0 u_0 = 1$$

$$\pi_0 = \frac{1}{2+2u_1-u_0}$$

At equilibrium, the distribution vector is obtained by solving the two equations

$$\mathbf{P}\,\mathbf{s} = \mathbf{s} \tag{10.68}$$

$$\sum \mathbf{s} = 1 \tag{10.69}$$

The solution to the above two equations is simple:

$$\mathbf{s} = \frac{1}{2 + u_1(n-1) - u_0} \begin{bmatrix} 1 \\ u_1 \\ u_1 \\ \vdots \\ u_1 \\ 1 - u_0 - u_1 \end{bmatrix} \tag{10.70}$$

CSMA/CD Protocol Performance

The throughput is given by the equation

$$\begin{aligned} \text{Th} &= \sum_{i=1}^{n} s_{t_i} \\ &= \frac{nu_1}{2 + u_1(n-1) - u_0} \end{aligned}$$

For large values of n, the throughput approaches 100% which is expected since little time is wasted during the collision.

Figure 10.7 shows the throughput of the IEEE 802.3 protocol when $n = 10$, $N = 10$, and $p = 1$. The solid line is the throughput of CSMA/CD, the dashed

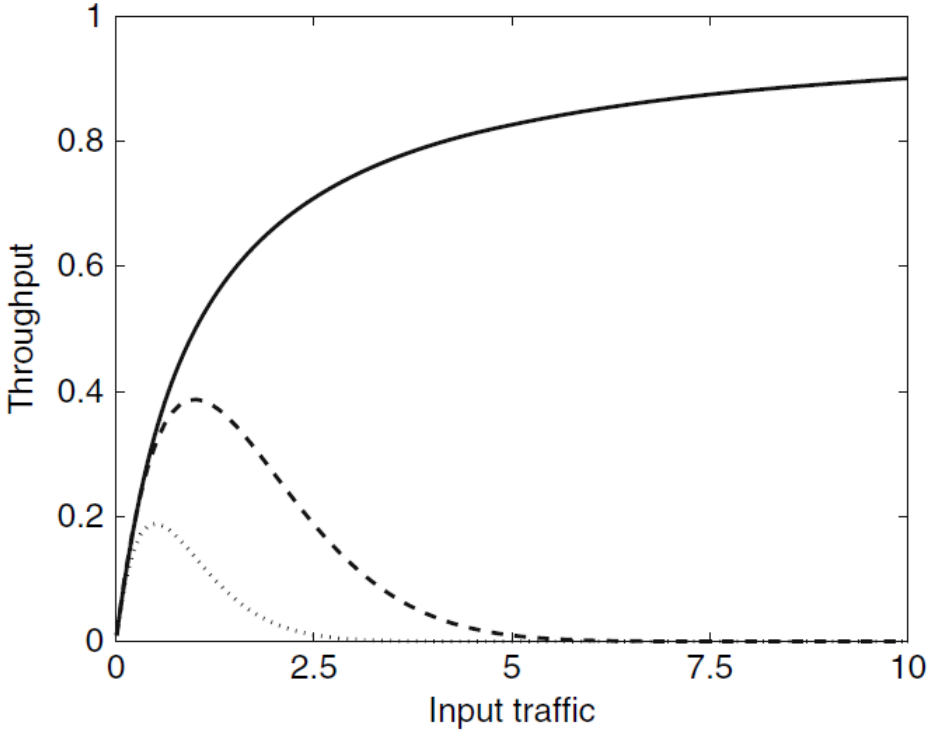


Fig. 10.7 The throughput for CSMA/CD versus the average input traffic when $n = 10$, $N = 10$, and $p = 1$. The *solid line* is the throughput of CSMA/CD, the *dashed line* represents the throughput of slotted ALOHA, and the *dotted line* represents the throughput of pure ALOHA

The access probability for the user in the CSMA/CD protocol is given by

$$p_a = \frac{\mathrm{Th}}{Na}$$

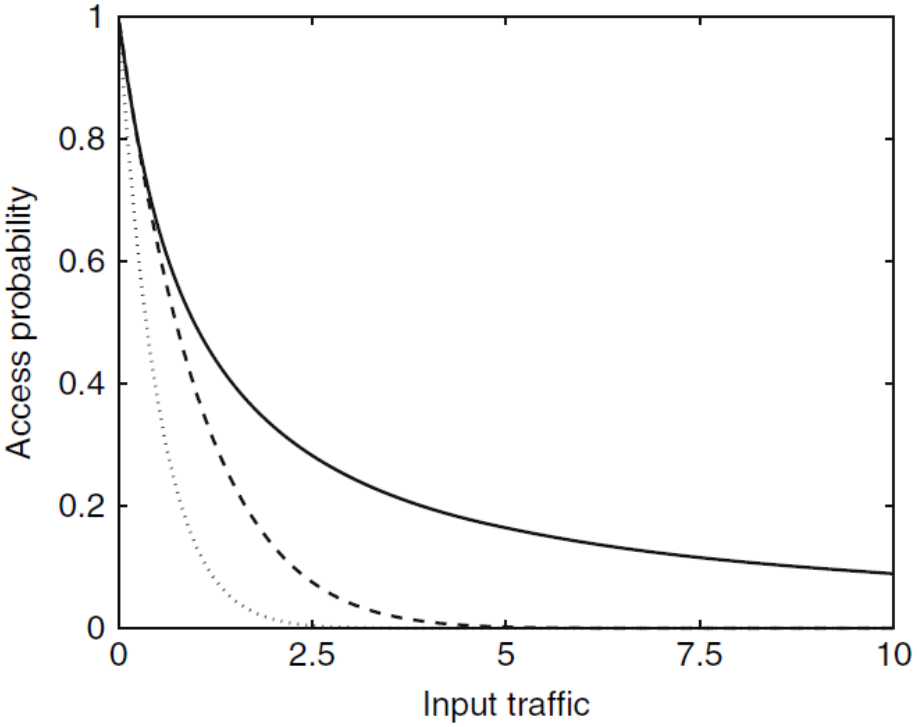


Fig. 10.8 Access probability of the IEEE 802.3 CSMA/CD protocol versus the average input traffic per time step for the case $n = 10$, $N = 10$, and $p = 1$. The *solid line* is the access probability of CSMA/CD, the *dashed line* represents the access probability of slotted ALOHA, and the *dotted line* represents the access probability of pure ALOHA

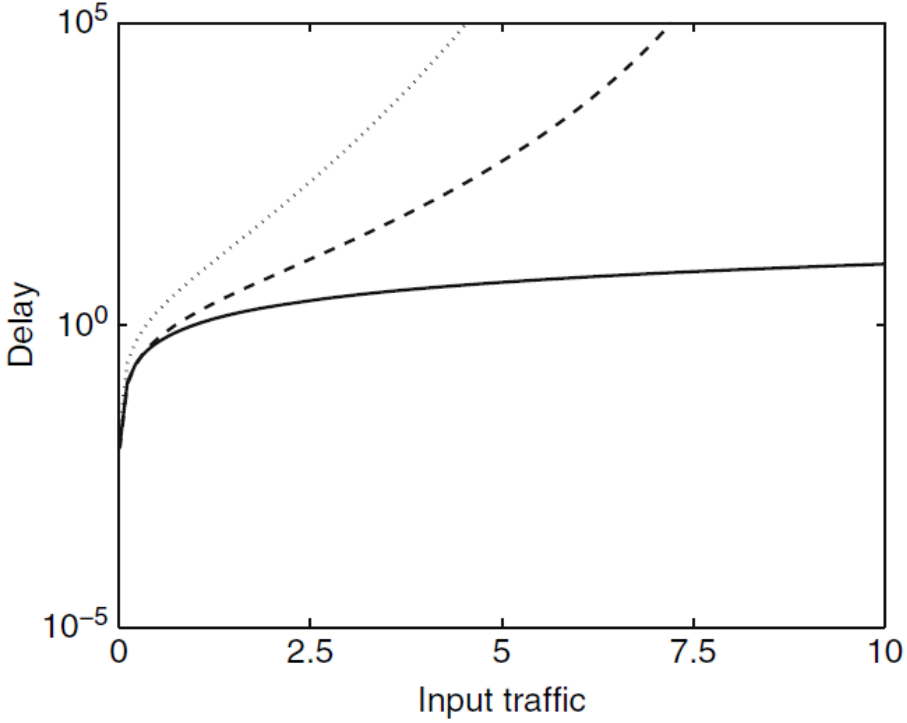


Fig. 10.9 Frame delay for the IEEE 802.3 CSMA/CD protocol versus the average input traffic frame for the case $n = 10$, $N = 10$, and $p = 1$. The *solid line* is the delay of CSMA/CD, the *dashed line* represents the delay of slotted ALOHA, and the *dotted line* represents the delay of pure ALOHA

The average number of attempts for a successful transmission is

$$n_a = \sum_{i=0}^{\infty} i\ (1 - p_a)^i\ p_a$$
$$= \frac{1 - p_a}{p_a}$$

Performance Analysis of CSMA/CA

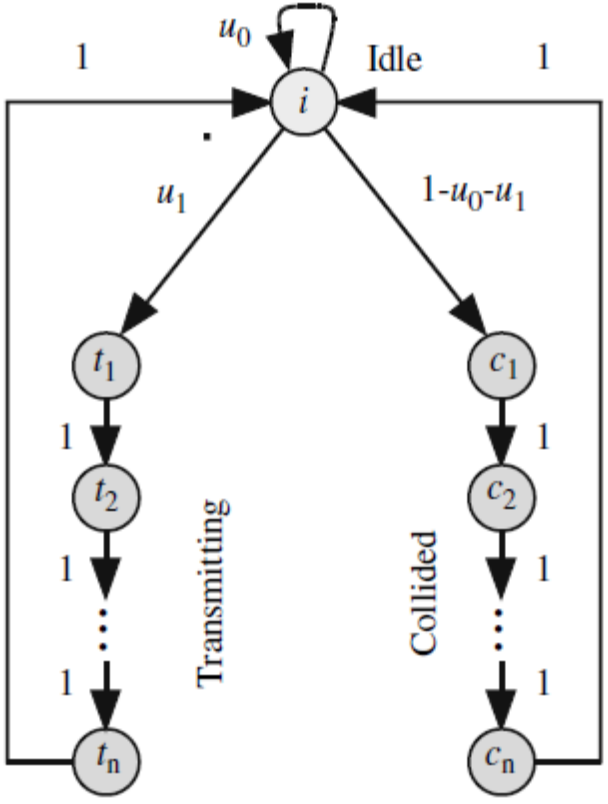
1. Since the current state of the channel depends only on its immediate past history, we can model the channel using Markov chain analysis.
2. The states of the Markov chain represent the states of the channel: *idle*, *transmitting*, and *collided*.
3. The channel is shared among N stations.
4. There is a single station class (equal priority).
5. The frame arrival probability per time step is a.
6. All transmitted frames have equal lengths.
7. A frame duration is equal to the transmission delay of a frame τ_t.
8. The time step of the Markov chain is chosen equal to the propagation delay, i.e., $T = \tau_p$.
9. We define n as the ratio of transmission delay to propagation delay, i.e., $n = \tau_t/\tau_p$. We assume that $n > 1$, which is true for small LANs carrying long frames or operating at low transmission speeds.
10. A 1-persistent CSMA/CA is assumed.
11. A station can have at most one message waiting for transmission.

The probability that i users are active at a given time step is given by

$$u_i = \binom{N}{i} a^i (1-a)^{N-i}$$

We organize the distribution vector at equilibrium as follows.

$$\mathbf{s} = \begin{bmatrix} s_i & s_{t_1} & s_{t_2} & \cdots & s_{t_n} & s_{c_1} & s_{c_2} & \cdots & s_{c_n} \end{bmatrix}^t$$


u_0
1
Idle
1
i
u_1
$1-u_0-u_1$
t_1
c_1
1
1
t_2
c_2
1
1
1
1
t_n
c_n
Transmitting
Collided

$$\mathbf{P} = \begin{bmatrix} u_0 & 0 & 0 & 1 & 0 & 0 & 1 \\ u_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 - u_0 - u_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{aligned} \mathbf{P}\,\mathbf{s} &= \mathbf{s} \\ \sum \mathbf{s} &= 1 \end{aligned}$$

$$\mathbf{s} = \frac{1}{n(1-u_0)+1}\begin{bmatrix} 1 \\ u_1 \\ u_1 \\ \vdots \\ u_1 \\ 1-u_0-u_1 \\ 1-u_0-u_1 \\ \vdots \\ 1-u_0-u_1 \end{bmatrix}$$

The throughput is given by the equation

$$\begin{aligned} \text{Th} &= \sum_{i=1}^{n} s_{t_i} \\ &= \frac{nu_1}{n(1-u_0)+1} \end{aligned}$$

For large values of N, we can write the throughput as

$$\text{Th} = \frac{n\mathrm{e}^{-\lambda}}{n\left(1-\mathrm{e}^{-\lambda}\right)+1}$$

For large values of n, the throughput approaches the expression

$$\begin{aligned} \text{Th} &\rightarrow u_1/(1-u_0) < 100\% \\ &\approx \frac{\lambda \mathrm{e}^{-\lambda}}{1-\mathrm{e}^{-\lambda}} = \frac{\lambda}{\mathrm{e}^{\lambda}-1} < 100\% \end{aligned}$$

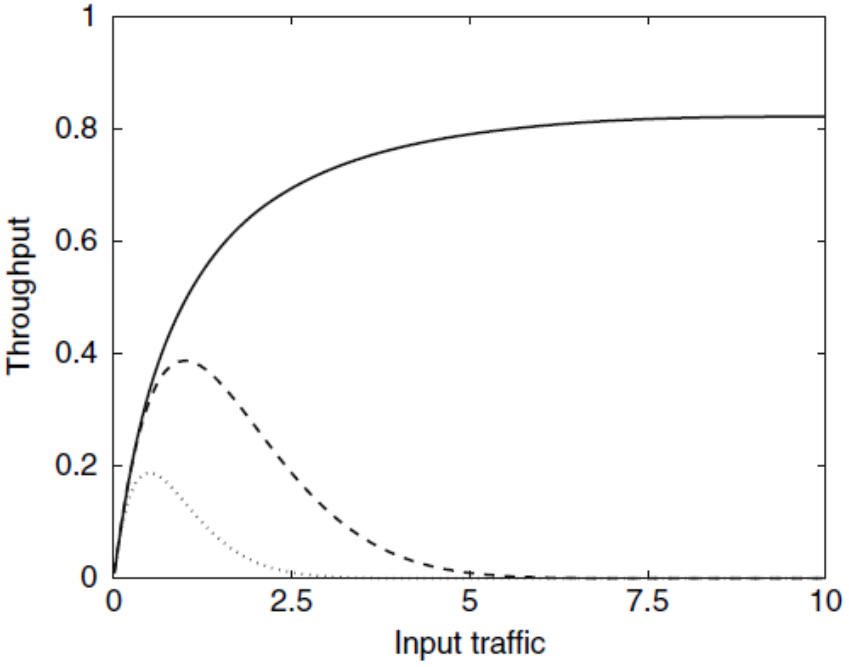


Fig. 10.11 The throughput for CSMA/CA versus the average input traffic when $n = 50$ and $N = 10$. The *solid line* is the throughput of CSMA/CA, the *dashed line* represents the throughput of slotted ALOHA, and the *dotted line* represents the throughput of pure ALOHA

The access probability for a user in the CSMA/CA protocol is given by

$$p_a = \frac{\mathrm{Th}}{Na}$$

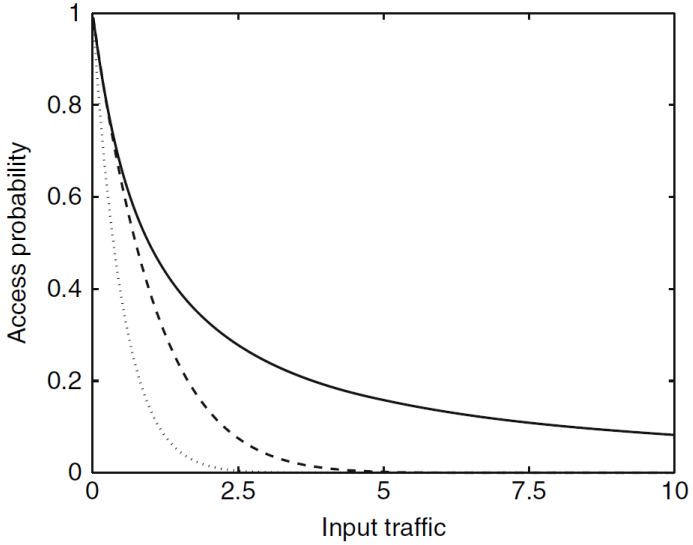


Fig. 10.12 Access probability of CSMA/CA protocol versus the average input traffic per time step for the case $n = 50$ and $N = 10$. The *solid line* is the access probability of CSMA/CA, the *dashed line* represents the access probability of slotted ALOHA, and the *dotted line* represents the access probability of pure ALOHA

The average number of attempts for a successful transmission is

$$
\begin{aligned}
n_a &= \sum_{i=0}^{\infty} i\ (1 - p_a)^i\ p_a \\
&= \frac{1 - p_a}{p_a}
\end{aligned}
$$

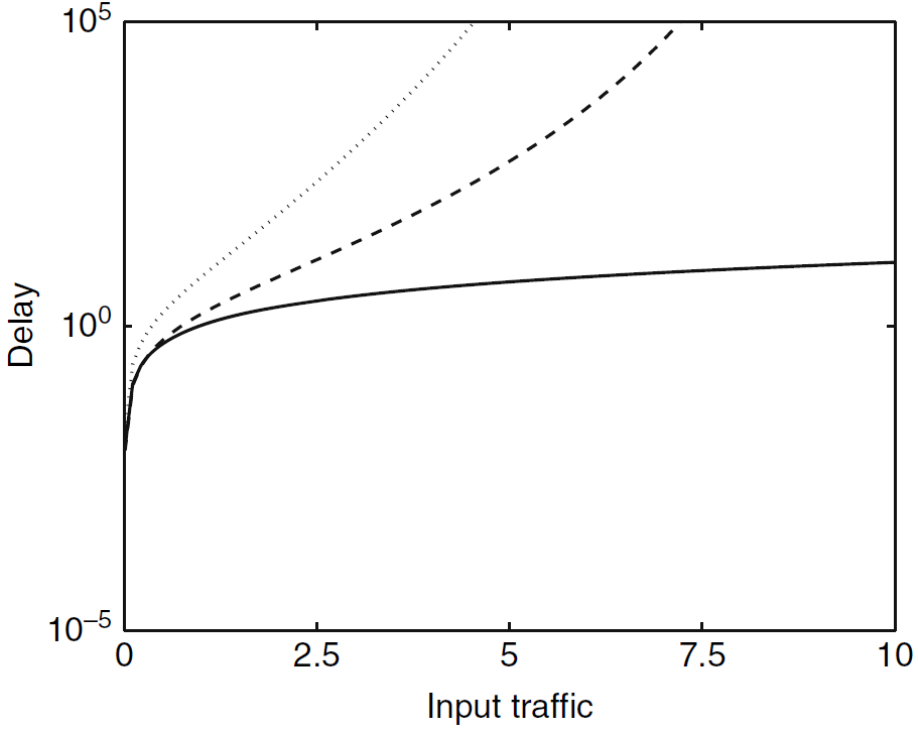
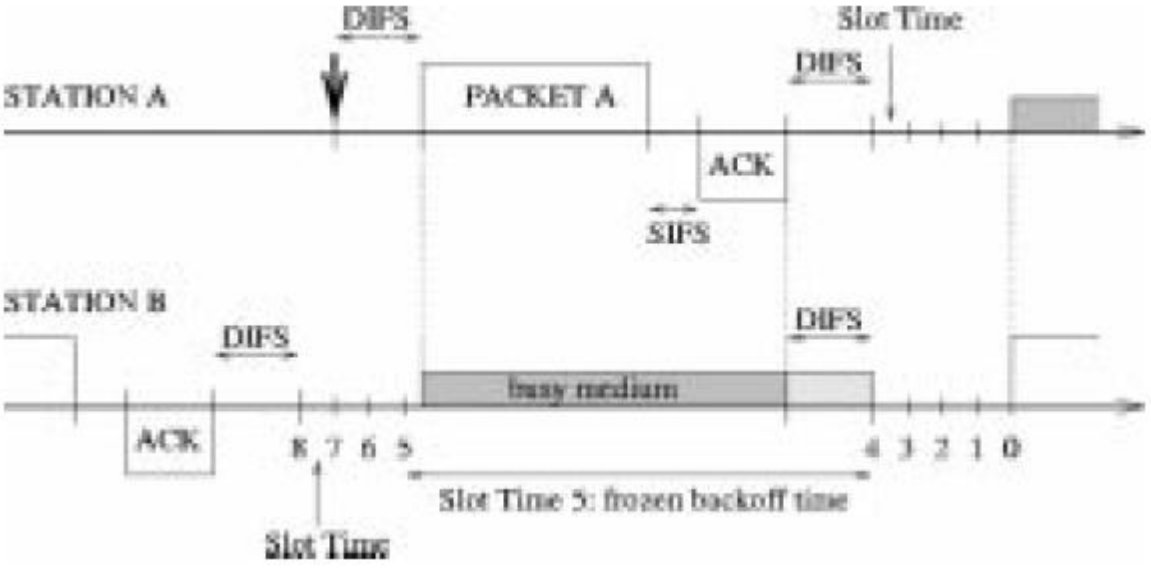


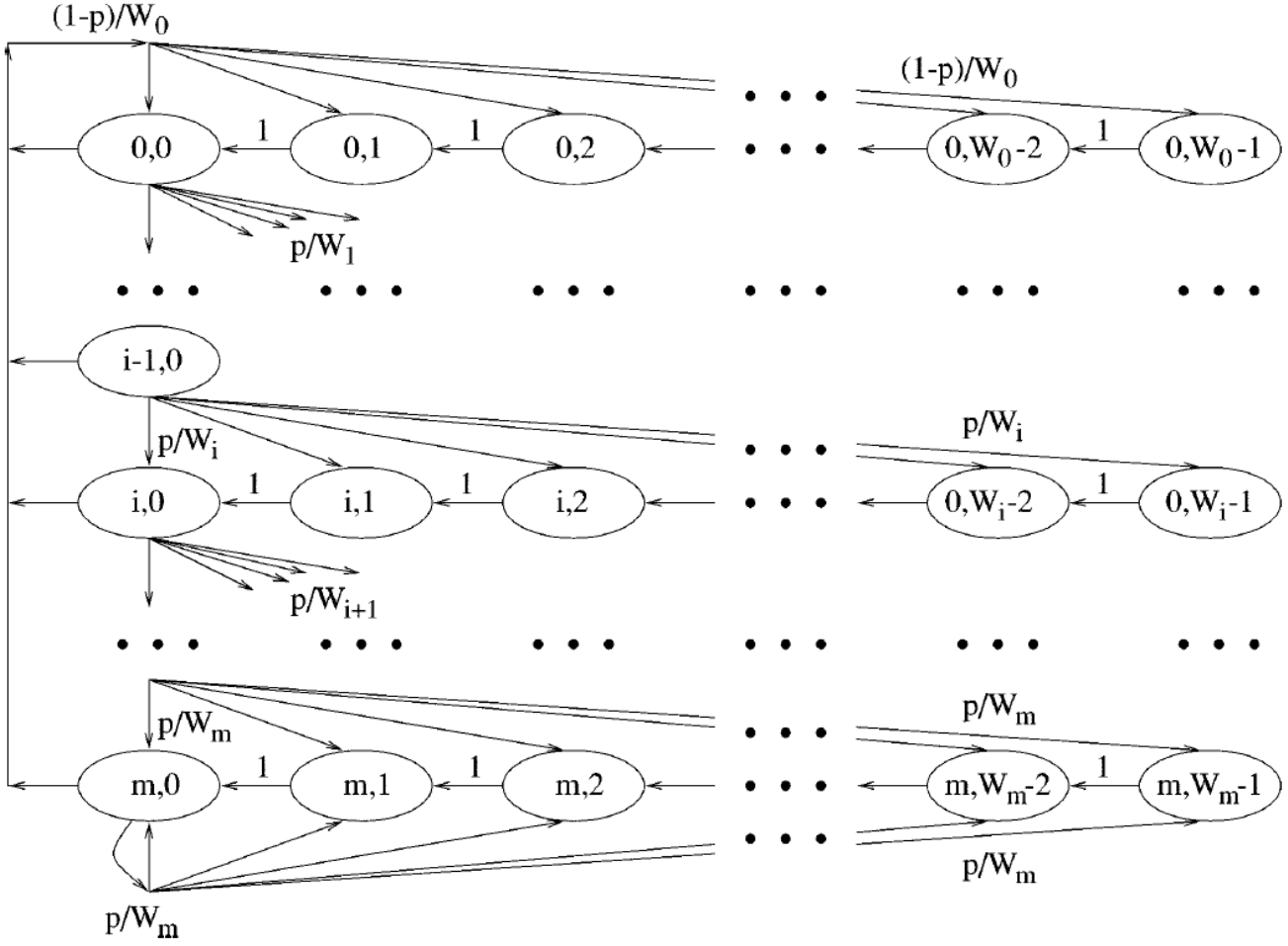
Fig. 10.14 Frame delay for the CSMA/CA protocol versus the average input traffic for the case $n = 50$ and $N = 10$. The *solid line* is the delay of CSMA/CA, the *dashed line* represents the delay of slotted ALOHA, and the *dotted line* represents the delay of pure ALOHA

802.11 DCF

Let $b(t)$ be the stochastic process representing the backoff time counter for a given station. A discrete and integer time scale is adopted: t and $t+1$ correspond to the beginning of two consecutive slot times, and the backoff time counter of each station decrements at the beginning of each slot time. Note that this discrete time scale does not directly relates to the system time. In fact, as illustrated in Fig. 1, the backoff time decrement is stopped when the channel is sensed busy, and thus the time interval between two consecutive slot time beginnings may be much longer than the slot time size σ, as it may include a packet transmission. In what follows, unless ambiguity occurs, with the term slot time we will refer to either the (constant) value σ, and the (variable) time interval between two consecutive backoff time counter decrements.


DIFS
Slot Time
STATION A
PACKET A
DIFS
ACK
SIFS
STATION B
DIFS
DIFS
busy medium
ACK
8 7 6 5
4 3 2 1 0
Slot Time 5: frozen backoff time
Slot Time

Since the value of the backoff counter of each station depends also on its transmission history (e.g., how many retransmission the head-of-line packet has suffered), the stochastic process $b(t)$ is non-Markovian. However, define for convenience $W = \mathrm{CW}_{\min}$. Let m, "maximum backoff stage," be the value such that $\mathrm{CW}_{\max} = 2^m W$, and let us adopt the notation $W_i = 2^i W$, where $i \in (0, m)$ is called "backoff stage." Let $s(t)$ be the stochastic process representing the backoff stage $(0, \ldots, m)$ of the station at time t.


(1-p)/W0
0,0
1
0,1
1
0,2
p/W1
(1-p)/W0
0,W0-2
1
0,W0-1
i-1,0
p/Wi
i,0
1
i,1
1
i,2
p/Wi+1
p/Wi
0,Wi-2
1
0,Wi-1
p/Wm
m,0
1
m,1
1
m,2
p/Wm
p/Wm
m,Wm-2
1
m,Wm-1
p/Wm

$$\begin{cases} P\{i,k \mid i,k+1\} = 1 & k \in (0, W_i - 2) \quad i \in (0,m) \\ P\{0,k \mid i,0\} = (1-p)/W_0 & k \in (0, W_0 - 1) \quad i \in (0,m) \\ P\{i,k \mid i-1,0\} = p/W_i & k \in (0, W_i - 1) \quad i \in (1,m) \\ P\{m,k \mid m,0\} = p/W_m & k \in (0, W_m - 1). \end{cases}$$

We can now express the probability τ that a station transmits in a randomly chosen slot time. As any transmission occurs when the backoff time counter is equal to zero, regardless of the backoff stage, it is

$$\tau = \sum_{i=0}^{m} b_{i,0} = \frac{b_{0,0}}{1-p} = \frac{2(1-2p)}{(1-2p)(W+1) + pW(1-(2p)^m)},$$

load bits. To compute S, let us analyze what can happen in a randomly chosen slot time. Let P_{tr} be the probability that there is at least one transmission in the considered slot time. Since n stations contend on the channel, and each transmits with probability τ

$$P_{\text{tr}} = 1 - (1 - \tau)^n. \tag{10}$$

The probability P_s that a transmission occurring on the channel is successful is given by the probability that exactly one station transmits on the channel, conditioned on the fact that at least one station transmits, i.e.,

$$P_s = \frac{n\tau(1-\tau)^{n-1}}{P_{\text{tr}}} = \frac{n\tau(1-\tau)^{n-1}}{1-(1-\tau)^n}. \tag{11}$$

We are now able to express S as the ratio

$$S = \frac{E\left[\text{payload information transmitted in a slot time}\right]}{E\left[\text{length of a slot time}\right]}. \tag{12}$$

Being $E[P]$ the average packet payload size, the average amount of payload information successfully transmitted in a slot time is $P_{\mathrm{tr}}P_s E[P]$, since a successful transmission occurs in a slot time with probability $P_{\mathrm{tr}}P_s$. The average length of a slot time is readily obtained considering that, with probability $1 - P_{\mathrm{tr}}$,

the slot time is empty; with probability $P_{\mathrm{tr}}P_s$ it contains a successful transmission, and with probability $P_{\mathrm{tr}}(1 - P_s)$ it contains a collision. Hence, (12) becomes

$$S = \frac{P_s P_{\mathrm{tr}} E[P]}{(1 - P_{\mathrm{tr}})\sigma + P_{\mathrm{tr}} P_s T_s + P_{\mathrm{tr}}(1 - P_s) T_c}.$$

There are three main types of ARQ techniques:

- Stop-and-wait ARQ (SW ARQ)
- Go-back-N ARQ (GBN ARQ)
- Selective-repeat ARQ (SR ARQ)

9.2 Stop-and-Wait ARQ (SW ARQ) Protocol

Stop-and-wait ARQ (SW ARQ) protocol is a simple protocol for handling frame transmission errors when the round-trip time ($2\tau_p$) for frame propagation and reception of acknowledgment is smaller than frame transmission time (τ_t). The propagation delay τ_p is given by

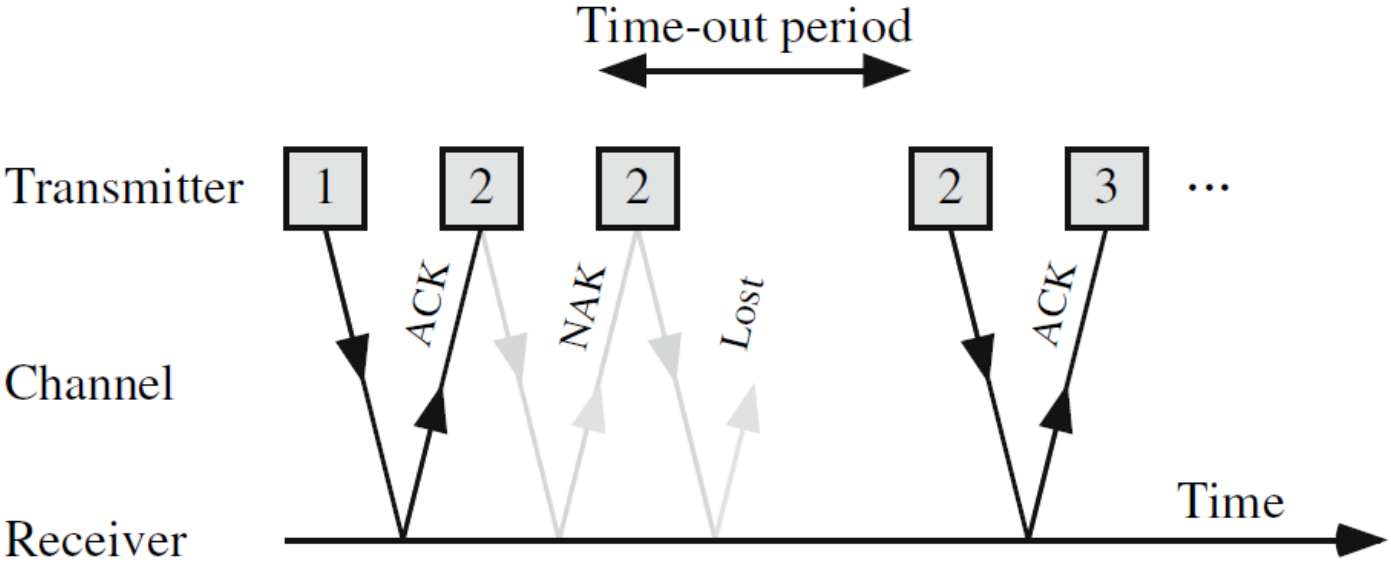
$$\tau_p = \frac{d}{c}$$

where c is speed of light and d is the distance between transmitter and receiver. The transmission delay τ_t is given by

$$\tau_t = \frac{L}{\lambda}$$

ARQ protocols are efficient and useful when we have

$$2\tau_p \ll \tau_t$$


Time-out period
Transmitter
Channel
Receiver
1
2
2
2
3
...
ACK
NAK
Lost
ACK
Time

9.2.1 Modeling Stop-and-Wait ARQ

1. The average length of a frame is n bits.
2. The forward channel has random noise and the probability that a bit will be received in error is ϵ. Another name for ϵ is bit error rate (BER) .
3. The feedback channel is assumed noise-free so that acknowledgment signals from the receiving station will always be transmitted to the sending station.
4. The sender will keep sending a frame until it is correctly received. The effect of limiting the number of retransmissions is discussed in Problem 9.2.

The state of the sender while attempting to transmit a frame depends only on the outcome of the frame just sent. Hence we can represent the state of the sender as a Markov chain having the following properties:

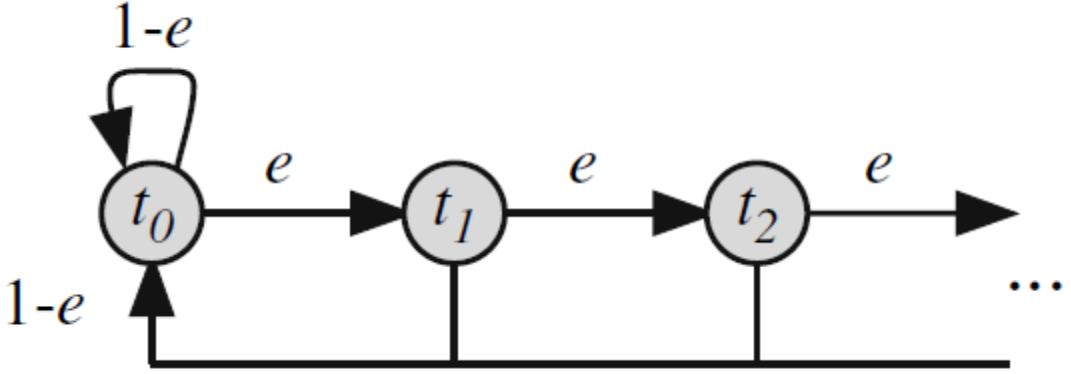
1. State i of the Markov chain indicates that the sender is retransmitting the frame for the ith time. State 0 indicates error-free transmission.
2. The number of states is infinite since no upper bound is placed on the number of retransmissions.
3. The time step is taken equal to the sum of transmission delay and round-trip delay $T = \tau_t + 2\tau_p$.

The state transition diagram for the SW ARQ protocol is shown in Fig. 9.2. In the figure, e represents the probability that the transmitted frame contained an error. e is given by the expression

$$e = 1 - (1 - \epsilon)^n \tag{9.2}$$

For a noise-free channel $\epsilon = 0$ and so $e = 0$. When the average number of errors in a frame is very small (i.e., $\epsilon\, n \ll 1$), we can write

$$e \approx \epsilon\, n \tag{9.3}$$


1-e
e
e
e
t_0
t_1
t_2
1-e
...

We organize the state distribution vector as follows:

$$\mathbf{s} = \begin{bmatrix} s_0 & s_1 & s_2 & \cdots \end{bmatrix}^t$$

$$\mathbf{P} = \begin{bmatrix} 1-e & 1-e & 1-e & 1-e & \cdots \\ e & 0 & 0 & 0 & \cdots \\ 0 & e & 0 & 0 & \cdots \\ 0 & 0 & e & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

At equilibrium, the distribution vector is obtained by solving

$$\mathbf{P}\,\mathbf{s} = \mathbf{s}$$
$$\sum \mathbf{s} = 1$$

The solution to the above two equations is simple:

$$\mathbf{s} = (1 - e) \times \left[1 \; e \; e^2 \; \cdots \right]^t$$

The average number of retransmissions for a frame is given by

$$
\begin{aligned}
N_t &= (1-e) \times (s_1 + 2s_2 + 3s_3 + \cdots) \\
&= (1-e)^2 \times \sum_{i=0}^{\infty} i\ e^i \\
&= e \qquad \text{transmissions/frame}
\end{aligned}
$$

For a typical channel, $e \approx \epsilon\, n \ll 1$ and the average number of transmissions can be approximated as

$$N_t \approx \epsilon\, n \tag{9.11}$$

We define the efficiency of the SW ARQ protocol as the inverse of the total number of transmissions which includes the first transmission plus the average number of retransmissions. In that case, η is given by

$$\eta = \frac{1}{1 + N_t} = \frac{1}{1 + e} \tag{9.12}$$

For an error-free channel, $N_t = 0$ and $\eta = 100\%$. For a typical channel, $e \approx \epsilon\, n \ll 1$ and the efficiency is given by

$$\eta \approx 1 - \epsilon\, n \tag{9.13}$$

The throughput of the transmitter can be expressed as

$$\text{Th} = \eta = 1 - e \qquad \text{frames/time step} \tag{9.14}$$

Thus, for an error-free channel, $\eta = 1$ and arriving frames are guaranteed to be transmitted on the first try. We could have obtained the above expression for the throughput by estimating the number of frames that are successfully transmitted in each transmitter state:

$$\begin{aligned} \text{Th} &= (1 - e) \sum_{i=0}^{\infty} s_i \\ &= 1 - e \end{aligned} \tag{9.15}$$

9.3 Go-Back-*N* (GBN ARQ) Protocol

In the go-back-*N* protocol, the transmitter keeps sending frames but keeps a copy in a buffer, which is called the transmission window. The number of frames in the buffer, or the window, is *N* which equals the number of frames sent during one round-trip time.

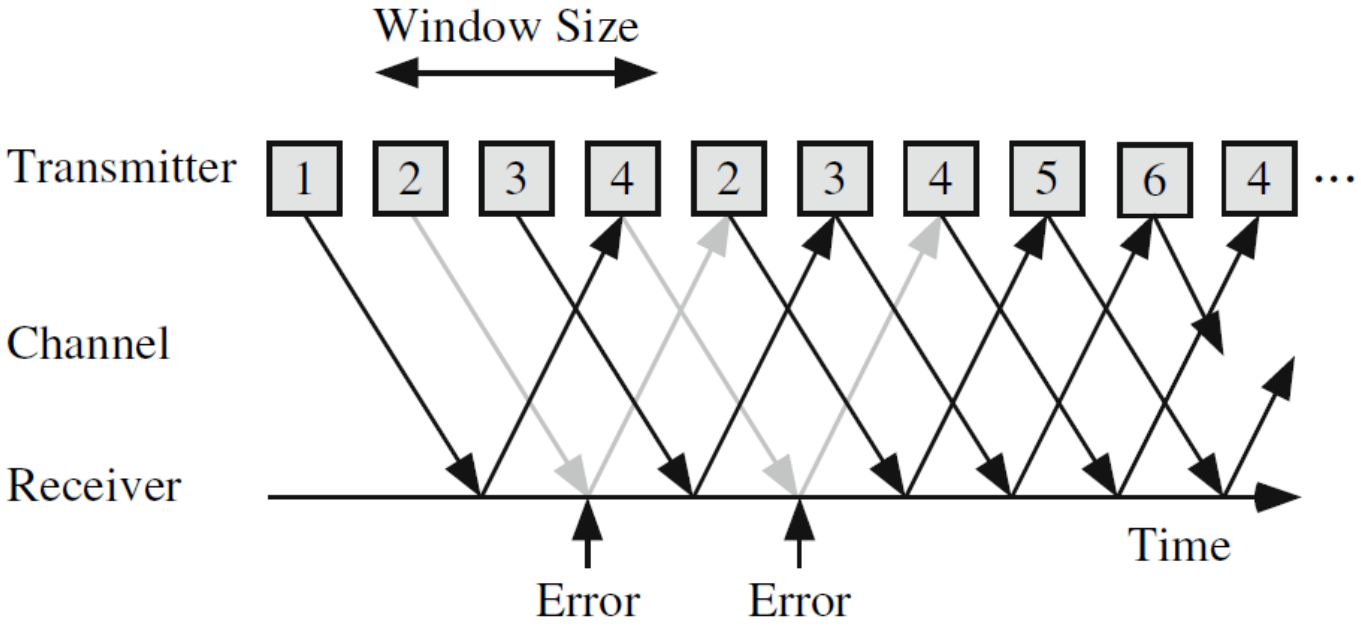


Fig. 9.3 Go-back-*N* ARQ protocol with buffer size *N* = 3. *Solid arrows* indicate ACK signals and *grey arrows* indicate NAK signals

9.3.1 Modeling the GBN ARQ Protocol

1. Each window contains N frames.
2. The average length of a frame is n bits.
3. The forward channel has random noise and the probability that a bit will be received in error is ϵ. Another name for ϵ is bit error rate (BER).
4. The feedback channel is assumed noise-free so that acknowledgment signals from the receiving station will always be transmitted to the sending station.
5. The maximum number of retransmissions is k_{m} after which the sender will declare the channel to be not functioning.

State of the Markov Chain

1. The states of the Markov chain are grouped into the sets $\mathcal{T}$, $\mathcal{R}_1$, $\mathcal{R}_2$, etc. These sets are explained below.
2. The number of states is infinite since no upper bound is placed on the number of retransmissions.
3. The time step is taken equal to the sum of transmission delay of one frame $T = \tau_t$. Thus, a window that contains N frames will require N time steps to be transmitted.

The set $\mathcal{T}$ represents the states of the sender while it is transmitting a window for freshly arrived N frames:

$$\mathcal{T} = \{ t_1 \; t_2 \cdots t_N \} \tag{9.16}$$

The set $\mathcal{R}_1$ represents the states of the sender while it is retransmitting frames for the first time due to a damaged or lost frame. This set is the union of several subsets

$$\mathcal{R}_1 = R_{1,1} \cup R_{1,2} \cup \cdots \cup R_{1,N} \tag{9.17}$$

where subset $R_{1,i}$ is the subset of $\mathcal{R}_1$ that contains i states corresponding to retransmitting i frames for the first time. In other words, the first $N - i$ frames have been correctly received. With the help of (9.23) below, we can verify that all states in subset $R_{1,i}$ are equal so that we can write the i states associated with $R_{1,i}$ as

$$R_{1,i} = \left\{ r_{1,i}\ r_{1,i}\ \cdots r_{1,i} \right\} \tag{9.18}$$

Thus $\mathcal{R}_1$ has N *unique* subsets such that subset $R_{1,i}$ has i equal states $r_{1,i}$:

$r_{1,1}$ corresponding to last frame in error

$r_{1,2}$ corresponding to frame before last in error

$\vdots$

$r_{1,N}$ corresponding first frame in error

Similarly, the set $\mathcal{R}_2$ represents the states of the sender while it is retransmitting frames for the second time. This set is the union of several subsets

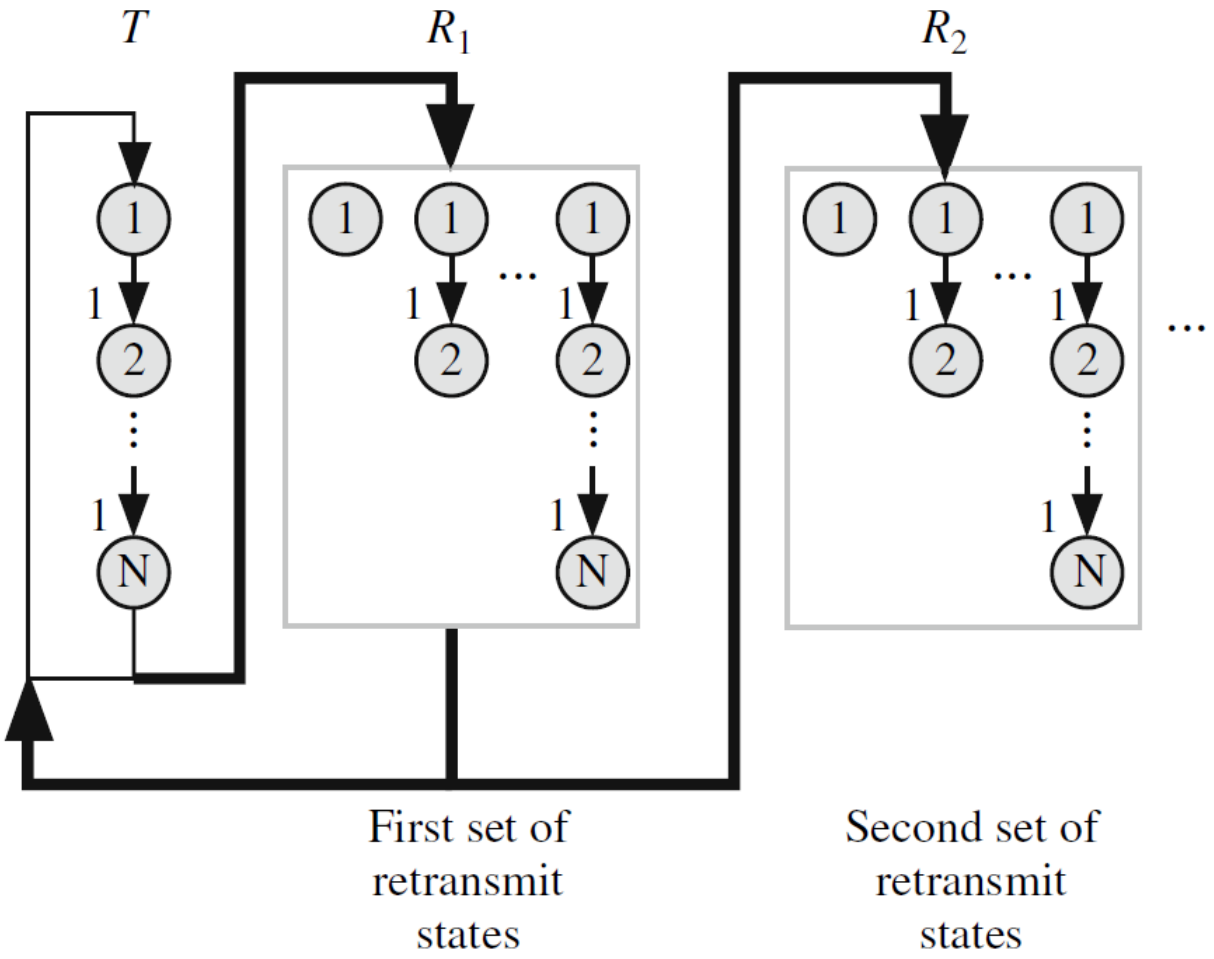
$$\mathcal{R}_2 = R_{2,1} \cup R_{2,2} \cup \cdots \cup R_{1,N} \tag{9.19}$$

where subset $R_{2,i}$ is the subset of $\mathcal{R}_2$ that contains i states corresponding to retransmitting i frames for the second time. With the help of (9.23), we can verify that all states in subset $R_{2,i}$ are equal so that we can write the i states associated with $R_{2,i}$ as

$$R_{2,i} = \{ r_{2,i}\ r_{2,i}\ \cdots\ r_{2,i} \} \tag{9.20}$$

Thus $\mathcal{R}_2$ has N *unique* subsets such that subset $R_{2,i}$ has i equal states $r_{2,i}$:

$r_{2,1}$ corresponding to last frame in error
$r_{2,2}$ corresponding to frame before last in error
$\vdots$
$r_{2,N}$ corresponding first frame in error


T
R1
R2
1
2
N
First set of retransmit states
Second set of retransmit states

We organize the state distribution vector in the following order:

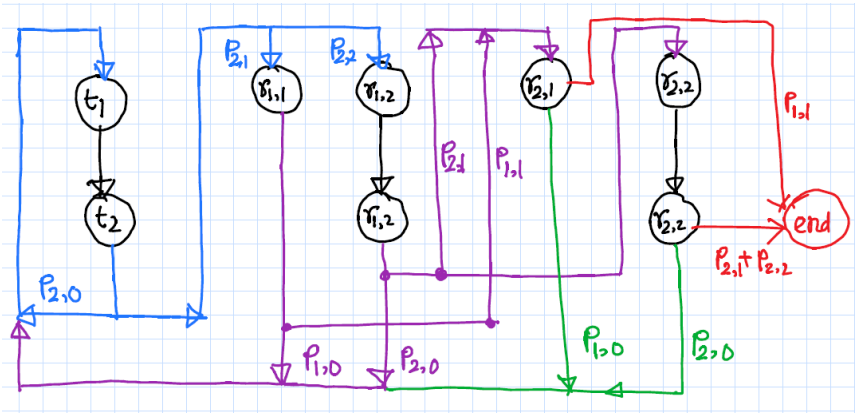
$$\mathbf{s} = \left[T \ \mathcal{R}_1 \ \mathcal{R}_2 \ \cdots \right]^t \tag{9.21}$$

For the case when $N = 2$, the distribution vector can be written as

$$\mathbf{s} = \left[\begin{array}{cc|ccc|ccc|c} t & t & r_{1,1} & r_{1,2} & r_{1,2} & r_{2,1} & r_{2,2} & r_{2,2} & \cdots \end{array} \right]^t \tag{9.22}$$

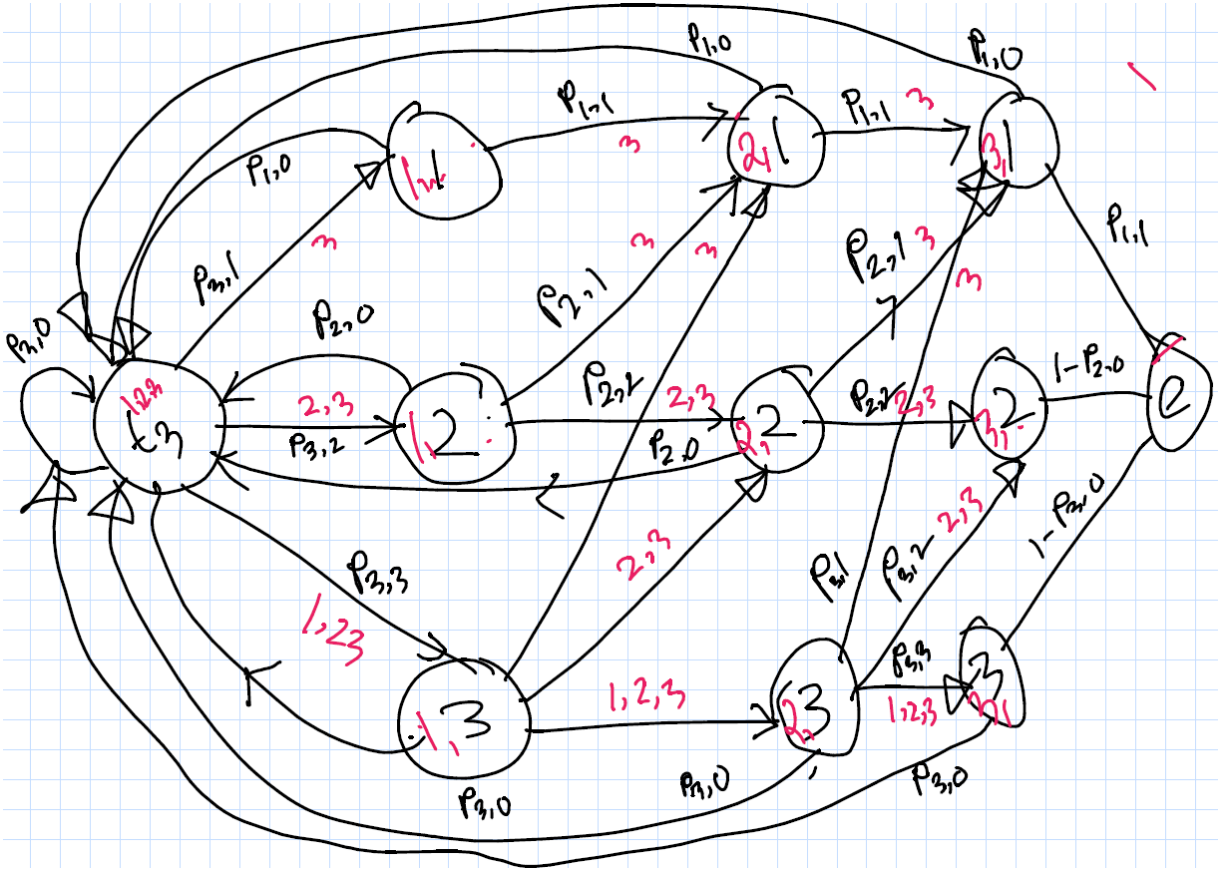
The corresponding transition matrix of the sender for the case when $N = 2$ is given by

$$\mathbf{P} = \begin{bmatrix} 0 & p_{2,0} & p_{1,0} & 0 & p_{2,0} & p_{1,0} & 0 & p_{2,0} & \cdots \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & p_{2,1} & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & p_{2,2} & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & p_{1,1} & 0 & p_{2,1} & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & p_{2,2} & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \quad (9.23)$$



$P =$

	t_1	t_2	$r_{1,1}$	$r_{1,2}$	$r_{1,2}$	$r_{2,1}$	$r_{2,2}$	$r_{2,2}$	...
t_1	0	1	0	0	0	0	0	0	
t_2	$p_{2,0}$	0	$p_{2,1}$	$p_{2,2}$	0	0	0	0	...
$r_{1,1}$	$p_{1,0}$	0	0	0	0	$p_{1,1}$	0	0	
$r_{1,2}$	0	0	0	0	1	0	0	0	...
$r_{1,2}$	$p_{2,0}$	0	0	0	0	$p_{2,1}$	$p_{2,2}$	0	
$r_{2,1}$	$p_{1,0}$	0	0	0	0	0	0	0	
$r_{2,2}$	0	0	0	0	0	0	0	1	
$r_{2,2}$	$p_{2,0}$	0	0	0	0	0	0	0	


1
$p_{1,0}$
$p_{1,1}$
$p_{2,0}$
$p_{2,1}$
$p_{2,2}$
$p_{3,0}$
$p_{3,1}$
$p_{3,2}$
$p_{3,3}$
$1-p_{2,0}$
$1-p_{3,0}$
1,2,3
2,3
3
0

In the above matrix, the transition probability $p_{i,j}$ is the probability that the last j frames need to be retransmitted, given that a frame of i frames was sent. For instance, $p_{5,2}$ indicates the probability that the last two frames have to be retransmitted, given that five frames were sent and the first three frames were received without error. $p_{i,j}$ is given by the expression

$$p_{i,j} = (1 - e)^{i-j} e \tag{9.24}$$

where e is the probability that a frame contained one or more errors:

$$e = 1 - (1 - \epsilon)^n \tag{9.25}$$

9.3.2 Using Iterations to Find π

we can easily prove that all transmit states in the set $\mathcal{T}$ are equal:

$$t = t_1 = t_2 = \cdots = t_N \tag{9.29}$$

In fact, all states in any column of the matrix are equal. For example, we can write the N *unique states* of $\mathcal{R}_1$ as

$$\begin{aligned} r_{1,N} &= t\ p_{N,N} && (9.30)\\ r_{1,N-1} &= t\ p_{N,N-1} && (9.31)\\ r_{1,N-2} &= t\ p_{N,N-2} && (9.32)\\ &\vdots \\ r_{1,1} &= t\ p_{N,1} && (9.33) \end{aligned}$$

In general, we have

$$r_{1,j} = t\ p_{N,j} \qquad j = 1,\ 2, \ldots,\ N$$

$$\begin{aligned}
r_{2,N} &= r_{1,N}\ p_{N,N} \\
r_{2,N-1} &= \sum_{i=N-1}^{N} r_{1,i}\ p_{i,N-1} \\
r_{2,N-2} &= \sum_{i=N-2}^{N} r_{1,i}\ p_{i,N-2} \\
&\vdots \\
r_{2,1} &= \sum_{i=1}^{N} r_{1,i}\ p_{i,1}
\end{aligned}$$

In general, we have

$$r_{2,j} = \sum_{i=j}^{N} r_{1,i}\ p_{i,j} \qquad\qquad j = 1,\ 2,\ \ldots,\ N$$

The states associated with the kth retransmission $\mathcal{R}_k$ are given by the iterative expression

$$r_{k,j} = \sum_{i=j}^{N} r_{k-1,i}\ p_{i,j} \qquad \begin{matrix} k = 2, 3, \ldots \\ j = 1, 2, \ldots, N \end{matrix} \tag{9.40}$$

with the initial condition

$$r_{1,j} = t\ p_{N,j} \qquad j = 1,\ 2,\ \cdots N \tag{9.41}$$

1. Assign to each transmit state some value, for example,

$$t = 1 \tag{9.42}$$

2. Estimate the retransmit states for $\mathcal{R}_1$ using the iterative expression (9.34).
3. Estimate the values of the other retransmit states $\mathcal{R}_k$ using the iterative expression (9.40).
4. Find the sum of all states

$$\mathcal{S} = N\, t + \sum_{k=1}^{k_m} \sum_{j=1}^{N} j\, r_{k,j} \tag{9.43}$$

5. The normalized value of the distribution vector is given by the following equation:

$$\mathbf{s} = \frac{1}{\mathcal{S}}\left[\begin{array}{cc|ccc|ccc|c} t & t & r_{1,1} & r_{1,2} & r_{1,2} & r_{2,1} & r_{2,2} & r_{2,2} & \cdots \end{array}\right]^t \tag{9.44}$$

$t = t_1 = t_2$

$r_{1,1} = t\, p_{2,1}$

$r_{1,2} = t\, p_{2,2}$

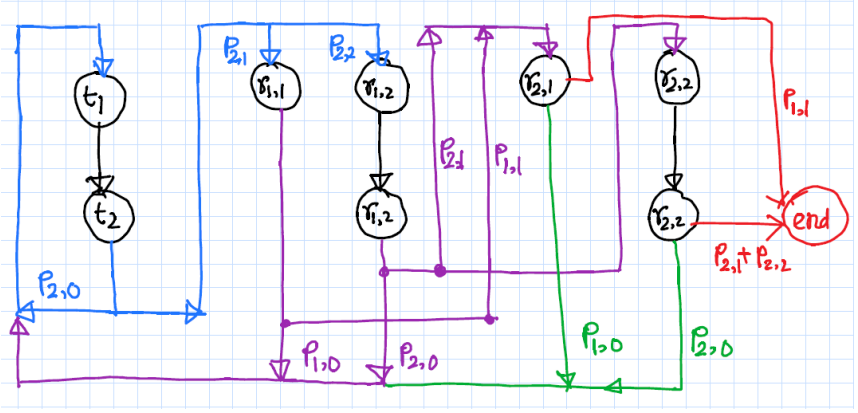
$r_{2,1} = r_{1,1}\, p_{1,1} + r_{1,2}\, p_{2,1}$

$r_{2,2} = r_{1,2}\, p_{2,2}$

$t_1 = r_{1,1}\, p_{1,0} + r_{1,2}\, p_{2,0} + r_{2,1}\, p_{1,0} + r_{2,2}\, p_{2,0}$

$e = r_{2,1}\, p_{1,1} + r_{2,2}\,(p_{2,1} + p_{2,2})$

$t + t_2 + r_{1,1} + 2\, r_{1,2} + r_{2,1} + 2\, r_{2,2} = 1$



9.3.4 GBN ARQ Performance

Estimating average number of retransmissions R_a

The probability that the source is in the kth retransmission state is given by

$$\alpha_k = \sum_{j=1}^{N} j \, r_{k,j} \tag{9.45}$$

α_k is also equal to the average number of frames sent at the kth retransmission (n_k).

The average number of retransmissions by the source for a given frame is given by

$$R_a = \sum_{k=1}^{k_m} k \, \alpha_k \tag{9.46}$$

Estimating average delay T_a

The delay associated with the kth retransmission is given by the accumulation of all the frames that were previously sent by earlier retransmissions. Thus, we can write

$$t_k = \sum_{j=1}^{k} \alpha_j \tag{9.47}$$

The average delay for transmitting a given frame is given by

$$T_a = \sum_{k=1}^{k_m} t_k \, \alpha_k \tag{9.48}$$

Estimating the average number of frames sent N_a

The average number of frames sent due to all retransmissions is given by

$$N_a = \sum_{k=1}^{k_m} n_k \, \alpha_k \tag{9.49}$$

GBN ARQ efficiency η and throughput (Th)

The efficiency of the GBN ARQ protocol is the ratio of frame size to the total number of frames transmitted:

$$\eta = \frac{N}{N + N_a} \tag{9.50}$$

When there are no errors, $N_a = 0$ and we get 100% efficiency.

9.4 Selective-Repeat (SR ARQ) Protocol

- Frame transmission errors when the round-trip time for frame transmission and reception of the acknowledgment is comparable to frame transmission time
- Used by the TCP transport protocol
- Each window contains *N* frames
- Receiver instructs the transmitter to resend only the frames that contained errors

9.4.1 Modeling the SR ARQ Protocol

1. Each window contains N frames.
2. The average length of a frame is n bits.
3. The forward channel has random noise and the probability that a bit will be received in error is ϵ. Another name for ϵ is bit error rate (BER).
4. The feedback channel is assumed noise-free so that acknowledgment signals from the receiving station will always be transmitted to the sending station.
5. The maximum number of retransmissions is k_{m} after which, the sender will declare the channel to be not functioning.

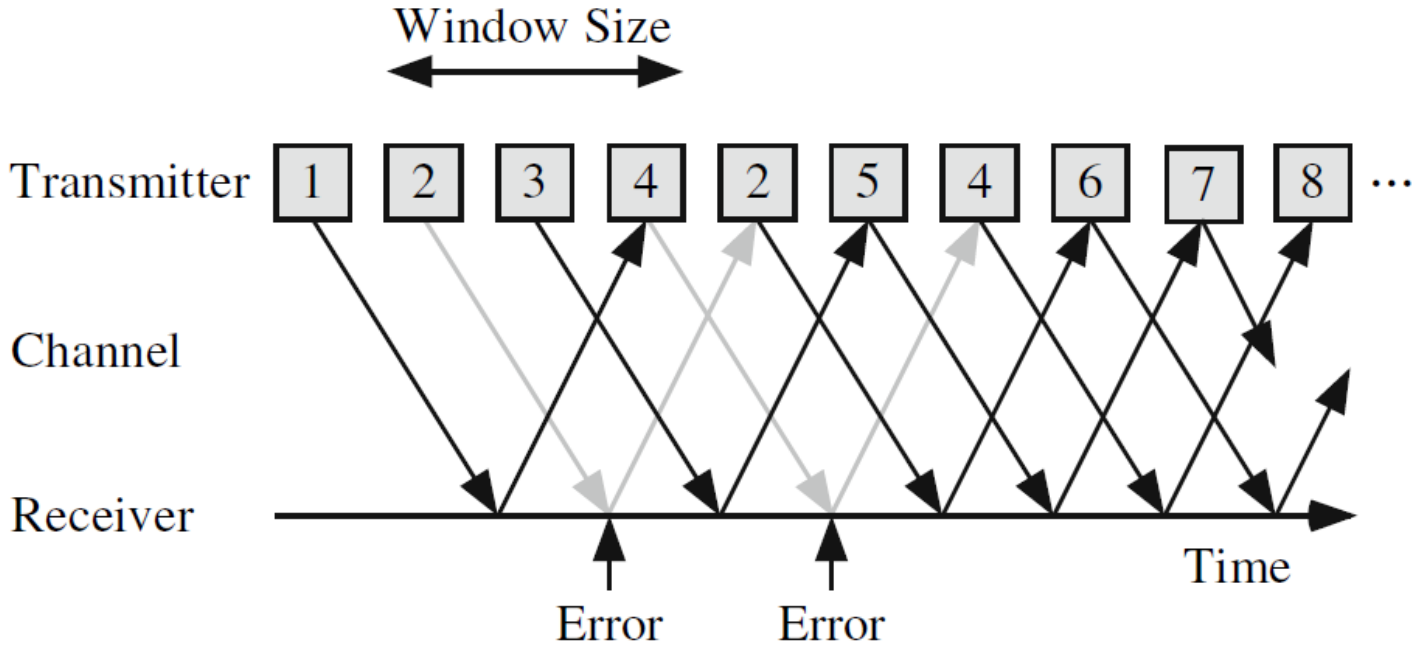


Fig. 9.5 Selective-repeat ARQ protocol with buffer size $N = 3$. *Solid arrows* indicate ACK signals and *grey arrows* indicate NAK signals

1. The states of the Markov chain are grouped into the sets $\mathcal{T}$, $\mathcal{R}_1$, $\mathcal{R}_2$, etc. These sets are explained below.
2. The number of states is infinite since no upper bound is placed on the number of retransmissions.
3. The time step is taken equal to the sum of transmission delay of one frame $T = \tau_t$. Thus, a window that contains N frames will require N time steps to be transmitted.

The set $\mathcal{T}$ represents the states of the sender while it is transmitting a window for freshly arrived N frames:

$$\mathcal{T} = \{ \; t_1 \quad t_2 \quad \cdots \quad t_N \; \} \tag{9.52}$$

The set $\mathcal{R}_1$ represents the states of the sender while it is retransmitting frames for the first time. This set is the union of several subsets

$$\mathcal{R}_1 = R_{1,1} \cup R_{1,2} \cup \cdots \cup R_{1,N} \tag{9.53}$$

where subset $R_{1,i}$ is the subset of $\mathcal{R}_1$ that contains i states corresponding to retransmitting i frames for the first time. With the help of (9.59), we can verify that all states in subset $R_{1,i}$ are equal so that we can write the i states associated with $R_{1,i}$ as

$$R_{1,i} = \left\{ \begin{array}{cccc} r_{1,i} & r_{1,i} & \cdots & r_{1,i} \end{array} \right\} \tag{9.54}$$

Thus $\mathcal{R}_1$ has N *unique* states:

$r_{1,1}$ repeated once,
$r_{1,2}$ repeated twice,
$\vdots$
$r_{1,N}$ repeated N times.

Similarly, the set $\mathcal{R}_2$ represents the states of the sender while it is retransmitting frames for the second time. This set is the union of several subsets

$$\mathcal{R}_2 = R_{2,1} \cup R_{2,2} \cup \cdots \cup R_{1,N} \tag{9.55}$$

where subset $R_{2,i}$ is the subset of $\mathcal{R}_2$ that contains i states corresponding to retransmitting i frames for the second time. With the help of (9.59), we can verify that all states in subset $R_{2,i}$ are equal so that we can write the i states associated with $R_{2,i}$ as

$$R_{2,i} = \{ \; r_{2,i} \quad r_{2,i} \quad \cdots \quad r_{2,i} \; \} \tag{9.56}$$

Thus $\mathcal{R}_2$ has N *unique* states:

$r_{2,1}$ repeated once,
$r_{2,2}$ repeated twice,
$\vdots$
$r_{2,N}$ repeated N times.

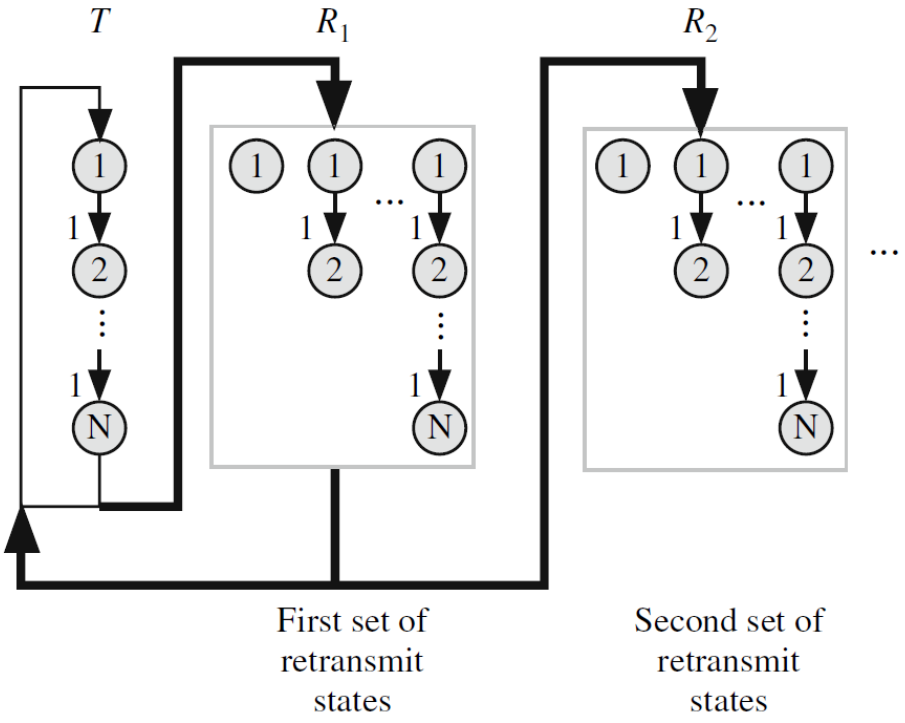


Fig. 9.6 State transition diagram of a sending station using the SR ARQ error control protocol. Only transitions in and out of R_1 are indicated

We organize the state distribution vector in the following order:

$$\mathbf{s} = \begin{bmatrix} T & \mathcal{R}_1 & \mathcal{R}_2 & \cdots \end{bmatrix}^t \tag{9.57}$$

For the case when $N = 2$, the distribution vector can be written as

$$\mathbf{s} = \begin{bmatrix} t & t & | & r_{1,1} & r_{1,2} & r_{1,2} & | & r_{2,1} & r_{2,2} & r_{2,2} & | & \cdots \end{bmatrix}^t \tag{9.58}$$

$$
\mathbf{P} = \begin{bmatrix}
0 & p_{2,0} & p_{1,0} & 0 & p_{2,0} & p_{1,0} & 0 & p_{2,0} & \cdots \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & p_{2,1} & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & p_{2,2} & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & 0 & p_{1,1} & 0 & p_{2,1} & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & p_{2,2} & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{bmatrix}
$$

In the above matrix, the transition probability $p_{i,j}$ is the probability that j frames need to be retransmitted, given that a frame of i frames was sent. For instance, $p_{5,2}$ indicates the probability that two frames have to be retransmitted, given that five frames were sent. $p_{i,j}$ is given by the expression

$$p_{i,j} = \binom{i}{j} (1-e)^{i-j}\, e^{j} \tag{9.60}$$

where e is the probability that a frame contained one or more errors:

$$e = 1 - (1-\epsilon)^{n} \tag{9.61}$$

At equilibrium, the distribution vector is obtained by solving the two equations

$$\mathbf{P}\,\mathbf{s} = \mathbf{s} \tag{9.62}$$

$$\sum \mathbf{s} = 1 \tag{9.63}$$

9.4.2 SR ARQ Performance

Estimating average number of retransmissions R_a

The probability that the source is in the kth retransmission state is given by

$$\alpha_k = \sum_{j=1}^{N} r_{k,j} \tag{9.65}$$

The average number of retransmissions by the source for a given frame is given by

$$R_a = \sum_{k=1}^{k_m} k\ \alpha_k \tag{9.66}$$

Estimating average delay T_a

The average number of frames sent at the kth retransmissions is given by

$$n_k = \sum_{j=1}^{N} j\ r_{k,j} \tag{9.67}$$

The delay associated with the kth retransmission is given by the accumulation of all the frames that were previously sent by earlier retransmissions. Thus, we can write

$$t_k = \sum_{j=1}^{k} n_j \tag{9.68}$$

The average delay for transmitting a given frame is given by

$$T_a = \sum_{k=1}^{k_m} t_k \, \alpha_k \tag{9.69}$$

Estimating the average number of frames sent N_a
The average number of frames sent due to all retransmissions is given by

$$N_a = \sum_{k=1}^{k_m} n_k \, \alpha_k \tag{9.70}$$

*SR ARQ efficiency η and throughput (*Th*)*

The efficiency of the SR ARQ protocol is the ratio of frame size to the total number of frames transmitted:

$$\eta = \frac{N}{N + N_{\mathrm{a}}} \tag{9.71}$$